

FOOD REMEDY API

RE005 – Research the Relationship Between Calories and Macronutrients in Food

Prepared For: Food Remedy API – Research Team

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1. Overview of the relationship: -

The calories provide the information about the value of the energy which the user is getting by eating the food. The user getting this energy is basically derived from three types of macro nutrients which are carbohydrates, proteins and the fats.

All three macro nutrients are very important as each macronutrient is essential for contributing towards the number of calories per gram in the particular product which helps to tell about the total energy content which is present in the particular food.

2. How actually the calories are derived from the macronutrients: -

The calories from the macro nutrients The human body plays the initial part of converting macro nutrients into energy by following processor of metabolism.

- The **Carbohydrates(Primary source of energy)** are broken into the glucose by the human body which is very essential for the primary source of energy to the body.
- For the first the body the **protein(Energy source which is used on Conditional purposes)** into amino acids which can be used for repairing the muscles and it also acted as an alternative energy source for the body.
- The **fats(High energy storing pathway)** are a little bit different because they're broken into fatty acids and the fat act as a long term reserve in perspective of energy that the body can in the long run if it is not getting appropriate level of carbohydrates and proteins.

So overall to find the total calories in to the particular product needs to be calculated by combining each contribution from different macro nutrients.



Carbohydrates (Primary Energy Pathway)

Carbohydrates are metabolised through:

- **Glycolysis** = glucose converting into the pyruvate
- **Citric Acid Cycle** (Krebs Cycle)
- **Electron Transport Chain** (ETC)

Proteins (Conditional Energy Source)

Proteins undergo:

- **Process of Deamination** (removal of nitrogen group from the body)
- Conversion into intermediates entering the **Krebs Cycle** which help to control the flow of citric acid in the body.

However:

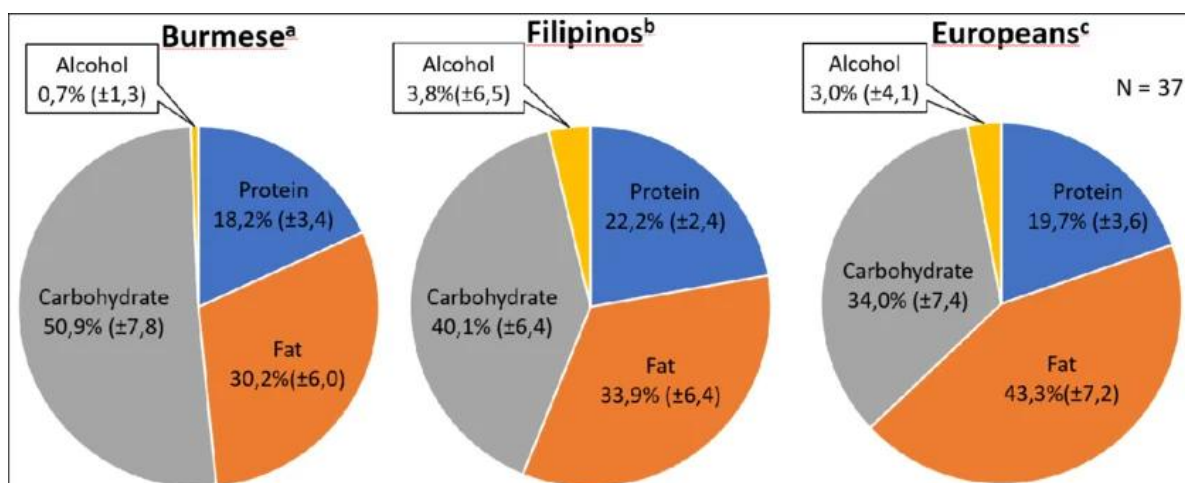
- This process is **less efficient** if the supply of carbohydrates is adequate.
- Produces metabolic waste (which is known as urea)
- Primarily used during **energy deficit**. Also when the intake of carbs are low and the circumstances when the people are fasting.

Fats (High-Energy Storage Pathway for long term)

Fats are metabolised through the:

- **Lipolysis** = it breakdown into fatty acids by following this process.
- **Beta-oxidation** = conversion of the fats into the acetyl-CoA
- Then the fats enter into the **Krebs Cycle** which is also known as the citric acid cycle.

Exp: 1 molecule of palmitic acid = 106 ATP



3. Defining values for calorie for each macronutrient: -

Macronutrients	Calories per gram	Human body process
Carbohydrates	4 kcal/g	Carbohydrates are the primary source of energy for the body.
Protein	4 kcal/g	With the help of protein the body becomes able to grow with the age and do the repairs related to tissues.
Fat	9 kcal/g	Fats can be used for long term storage as a energy reserve and it in the Insulation as well

4. The formula which is used for calculating the calories: -

To find out the total content of calories into the particular food the formula can be used:

$$\text{Total Calories} = (\text{Carbohydrates} \times 4) + (\text{Protein} \times 4) + (\text{Fat} \times 9)$$

This is the formula which is used by different types of labelling companies so that they've become able to find out the total calorie content into the food Multiplying carbohydrates with, protein with 4 as well and fat with 9.

5. Applying the formula on different products: -

Example 1: Banana

- Carbohydrates: $27\text{g} = 27 \times 4 = 108 \text{ kcal}$
- Protein: $1\text{g} = 1 \times 4 = 4 \text{ kcal}$
- Fat: $0.3\text{g} = 0.3 \times 9 = 3 \text{ kcal}$

Total Calories = 115 kcal

Example 2: Boiled Egg

- Carbohydrates: $1\text{g} = 4 \text{ kcal}$
- Protein: $6\text{g} = 24 \text{ kcal}$
- Fat: $5\text{g} = 45 \text{ kcal}$

Total Calories = 73 kcal

Example 3: = Burger

- Carbohydrates: $30\text{g} = 120 \text{ kcal}$
- Protein: $15\text{g} = 60 \text{ kcal}$
- Fat: $20\text{g} = 180 \text{ kcal}$

Total Calories = 360 kcal

Example 4: White Rice

- Carbohydrates: 45g = 180 kcal
- Protein: 4g = 16 kcal
- Fat: 0.5g = 4.5 kcal

Total Calories = 200 kcal

6. Conclusion: -

To conclude I want to mention to find the calorie content in any food the user must need to check the total composition of the micronutrients into that food. According to the research that I did the facts contribute highest amount energy program so that it helps increase the intake of the calorie. By doing the analysis of carbohydrates, proteins and fats I found that the user can easily estimate amount of energy they are int by eating the particular food item.

7. References: -

- Whitney, E. & Rolfes, S.R., 2020, Understanding Nutrition, Cengage Learning.
- National Health and Medical Research Council (NHMRC), 2013, Australian Dietary Guidelines.
- Food Standards Australia New Zealand (FSANZ), 2024, Nutrition Information Requirements.